Advertising in Toxic Feeds: Ad Effectiveness & Brand Safety

Registered Report Stage 1 Protocol

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Toxic content on social media captures attention and drives engagement, yet brands overwhelmingly treat it as a reputational threat and withdraw advertising from toxic environments. These two responses reflect two literatures that examine the effects of toxicity in isolation: brand safety research documents associative contamination from unsafe adjacent content, while platform engagement research documents attentional and behavioral gains from negative and toxic content. No existing work examines both within the same feed exposure. We integrate these perspectives in a single empirical framework. Using a dose-response design with stimulus sampling, we manipulate the proportion of toxic content in simulated social media feeds and measure feed-level engagement, ad-level attention, and brand-level reputational outcomes. We predict an inverted-U relationship between feed-level toxicity and both feed engagement and ad engagement: moderate toxicity elevates arousal and deepens processing; high toxicity triggers aversion and disengagement. The reputational effect of toxicity on brand attitude is theoretically ambiguous — the same attentional gains that benefit the ad also increase exposure to the aversive context — and is examined as a research question. Our design independently varies feed-level toxicity dose and adjacent-post toxicity, allowing simultaneous estimation of global and local effects. The findings inform brand safety strategy by providing a basis for evaluating the net effect of feed-level toxicity on advertising effectiveness.

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Words: 4837

Introduction

Social media platforms have increasingly become home to toxic discourses. Harassment, hate speech, cyberbullying, identity attacks, and deliberately insulting language pervade feeds across topics and languages, driven by recommendation algorithms and virality mechanisms amplifying such content. This toxicity poses a challenge for the multi-billion dollar advertising industry operating on these platforms. When toxicity on X (formerly Twitter) escalated following changes in ownership and content moderation in late 2022, major advertisers including Apple, Disney, and IBM suspended their campaigns; the platform's ad revenue declined by more than half within a year (PLACEHOLDER: Insider Intelligence 2023 forecast; confirm exact citation). The episode was not isolated. The World Bank pulled advertising from X in 2024 after a paid placement appeared alongside content from a white supremacist organization (PLACEHOLDER: CBS/WSJ report citation). These incidents foreground a tension that confronts brands that advertise in social media: the same platforms that reach billions of consumers also expose branded content to environments brands cannot fully control. With global social media advertising spend exceeding $275 billion in 2025 (Statista 2025) and social channels attracting the largest share of incremental ad dollars worldwide (WARC 2025), the reputational and financial exposure and risk to brands might be substantial.

Marketing research examines this risk to brands through the lens of brand safety. Recent work in this tradition documents that the proximity of branded content to unsafe or negative material can contaminate brand perceptions, even when no explicit link between the brand and the adjacent content exists (Grewal, Vana, and Stephen 2025). The general recommendation, followed by both theory and practice, is that any association with this type of content should be avoided. Accordingly, brand safety strategies, for example detection algorithms or moderation, have become a major point of concern for both brands and platforms eager to avoid any unsafe content association. With regard to unsafe content, toxicity may represent a particularly potent source of such contamination, given its harmful and targeted nature. Yet, while brand safety concerns are real and demonstrably shape advertiser behavior, the actual reputational spillover may be weaker under naturalistic feed conditions than this experimental literature suggests. Prior designs typically isolate a single brand-content pairing, a setup that likely inflates contamination effects relative to the cluttered, high-velocity environment of an actual social media feed. Recent null findings are consistent with this concern (Bernritter et al. 2025).

Interestingly, a separate stream of research on platforms puts forward the idea that negative and toxic content can have positive implications (CITE). Negatively-valenced content, and toxic content specifically, captures disproportionate attention and generates higher engagement on digital platforms driven by algorithms and virality mechanisms (Beknazar-Yuzbashev et al. 2022;

ADD). The platform-level implication seems to be: toxic content retains and engages users. From this perspective, feeds with higher toxicity might expose co-located advertisements to larger and more engaged audiences.

These two streams generate an interesting context for the same managerial question: Should my brand withdraw from toxic discourse or not? Brand safety research advises withdrawal from toxic environments due to potential spillovers. Engagement research implies that those same environments could increase exposure, attention, and engagement. We argue that existing work cannot resolve this contradiction because each stream examines its mechanism in isolation. Many brand safety studies measure reputational or attitudinal outcomes without accounting for the attention environment that toxicity creates. Platform engagement studies measure platform-level metrics without tracking whether associative contamination offsets the attentional gains. Marketing research has yet to examine both within the same feed exposure, leaving brands and platforms without a basis for evaluating the net effect of feed-level and adjacent toxicity on advertising effectiveness.

We address this gap by integrating brand safety and platform engagement within a single empirical framework. Focusing on digital advertising on social media, where toxicity is widespread and brands exercise little control over adjacent content, we examine whether the relationship between feed-level toxicity and brand advertising effectiveness is more complex than either stream in isolation suggests. We manipulate the proportion of toxic content in simulated but more naturalistic social media feeds using a dose-response design (with stimulus sampling CITE) and measure both brand-directed (click-through rate, ad recall, brand attitude) and feed-level outcomes (dwell time, time spent on feed, user reactions).

Our central argument is that toxicity operates through two mechanisms. At moderate levels, it elevates feed-level engagement and thereby increases the probability that co-located ads receive attention, producing net-positive effects on engagement and attention. At high levels, the attentional benefit saturates and feed-level toxicity yields net-negative effects for the ad due to decreasing engagement with the feed (or a spillover at extreme levels?) Taken together, we expect the resulting relationship between feed-level toxicity and brand advertising effectiveness follows an inverted U. At the same time, direct reputational spillover from toxic content to the co-located brand is absent across most toxicity levels that are not extreme. Reputational spillover may be weaker under naturalistic feed conditions than prior experimental designs — which typically isolate a single brand-content pairing — would suggest, a possibility consistent with recent null findings (Bernritter et al. 2025). The engagement benefits documented in the opposing stream are real but bounded: they hold up to a threshold beyond which the feed environment turns hostile enough to suppress even the attentional gains.

Theory

Toxicity in Online Discourse

Toxicity describes intentionally inflammatory communication directed at a target to provoke unproductive conflict (Hanscom et al. 2024). It subsumes harassment, hate speech, cyberbullying, identity attacks, profanity, and threats, unified by a common function of inflicting harm on a target (Hanscom et al. 2024; Lees et al. 2022; Shahi and Majchrzak 2025). Crucially, toxicity is a property of communicative style rather than subject matter as it characterises how a message is deployed irrespective of the topic it addresses. Toxicity's purpose is to damage rather than to evaluate, criticize, or persuade per se. Constructive criticism, passionate disagreement, and even harsh evaluative language can be productive. Toxicity has no regard for disagreement or advancing an argument.

Because toxicity is defined by communicative function rather than topic, it follows that toxic discourse is pervasive across platforms, languages, and topics (Avalle et al. 2024; Shahi and Majchrzak 2025). It emerges wherever users interact, not only around highly controversial topics, such as politics, but also in entertainment, sports, and other consumer contexts (Salminen et al. 2021; Tufa, Markov, and Possen 2024). Platform architectures amplify this prevalence through recommendation algorithms, virality features, and network effects (Recuero 2024; Hanscom et al. 2024). Toxicity's consequences extend beyond the immediate online interaction. For instance, offline, inflammatory online content is linked to political unrest and hate crime (Müller and Schwarz 2021, 2023). For platforms, toxicity generates regulatory pressure: legislation in the EU and elsewhere mandates moderation of toxic content (Digital Services Act 2022; UK Online Safety Act 2023).

This definition also implies that toxicity is distinct from negativity as a theoretical construct.. Negativity is valence-defined: it encompasses any communication carrying negative sentiment, including product complaints, critical reviews, and expressions of dissatisfaction. These forms of negative communication can serve evaluative, corrective, or persuasive functions and are vital components of healthy discourse. Toxicity isolates the subset of negative communication that is specifically target-directed and harm-oriented. This distinction matters for marketing and advertising outcomes because the mechanisms differ. Negativity affects mood and may produce valence-based affective spillover (Bushman 2005; de Pelsmacker et al. 2002). Toxicity additionally activates threat detection and increases arousal that might heighten attention or even produce aversion at high levels (Öhman and Mineka 2001; Lenhart et al. 2016). An examination

grounded in negativity alone cannot account for these different mechanisms.

Table 1. The two-dimensional nature of toxic communication. Valence (positive/negative) and communicative style (civil/toxic) are orthogonal: toxic communication is defined by its harm-oriented function, not by its sentiment.

	Civil	Toxic
Positive	"Great game today, the team really pulled through in the second half."	"So proud of this squad. The fans rooting against us are pathetic and deserve to suffer."
Negative	"Disappointing result today. The defense looked completely disorganized and the midfield offered nothing."	"You're a disgrace to this club. Hope you break your leg next match, you useless idiot."

The same definition clarifies why toxicity differs from the content categories that govern programmatic brand safety practice and offers a complementary perspective for brand safety. Established industry standards or individual company solutions implement blocklists or exclusion checklists that classify content by topic: violence, adult material, hate speech, and similar categories (Amazon Ads, 2025; GARM, 2024[1]). Keyword blocklists implement these classifications by flagging topical terms regardless of communicative intent or intensity. These operationalizations are coarse and lead to type-1 and type-2 errors: On the one hand, keyword blocklists most likely fail to meet their goal whenever a discussion on a harmless topic turns toxic. A discussion on sports competitions, while generally considered a harmless topic, might turn toxic via its communicative style. On the other hand, these same blocklists might restrict advertising in harmless discussions for flagging keywords that are harmful only in specific contexts. These limitations are well documented. As an advertising executive noted: a "keyword blocklist approach frequently results in inventory that is contextually relevant, and safe, being blocked." For example, during the Euro 2024 final, the flagging of keywords, such as "shoot" or "attack" led to nearly half of a UK publisher's output being blocked from receiving advertising (The Media Leader 2024). In comparison, toxicity captures *how* users communicate, which is orthogonal to what these users communicate about. Relying on toxicity for brand safety might thus flag unsafe contexts with higher specificity (reducing over-blocking of safe inventory tied to type-1 errors) and precision (improving detection of toxic discussions on nominally harmless topics that topic-based approaches miss). At the same time, the captured dimension of communication style is what regularly produces attentional, attitudinal, and arousal effects relevant to advertising outcomes (see "Motivating Evidence" section).

[1]While GARM was discontinued in 2024, its underlying content categorization remains broadly in use (Meier, 2025).

Toxicity and Advertising

We examine two brand outcomes under feed-level toxicity. First, we evaluate ad effectiveness through (in)attention, which we approximate through unobtrusively measured behavioral measures (dwell time, clickthrough rate) and survey responses on memory (ad recall). Second, we investigate classic brand safety measures conceptualized as whether the advertising context harms the advertised brand. We measure these through survey responses on attitudinal indicators (brand attitude, brand trust). These outcome classes require separate theoretical treatment because they operate through different mechanisms.

Ad Effectiveness

Toxicity drives both attention and engagement. Negatively-valenced content in general captures disproportionate attention (Pratto and John 1991; Rozin and Royzman 2001). It receives higher engagement rates, greater sharing, and stronger recall than neutral and positive content (for an overview, see Robertson et al. 2023). Further, threat-relevant stimuli, such as toxic communication (e.g., personal insult), activate automatic detection processes that elevate attention (Öhman and Mineka 2001). Users who face the ad may process it shallowly or deeply depending on their arousal state at the moment of encounter. Excitation transfer theory holds that residual arousal from a preceding stimulus intensifies affective and cognitive responses to subsequent stimuli (Zillmann 1971). In a social media feed, toxic posts preceding the ad elevate the user's arousal through their provocative and targeted content. This residual arousal transfers to the ad. With toxicity, elevated arousal deepens processing of the ad.

Toxic content, specifically, has been demonstrated to generate larger, wider, and deeper reply threads than civil counterparts on Twitter (Saveski et al. 2021) and across other platforms (Avalle et al. 2024). The other way around, reducing toxicity on a platform via a browser extension showed a decrease in user attention and engagement across time spent, impressions, and clicks (Beknazar-Yuzbashev et al. 2025) in line with this reasoning. The provocative and targeted nature of toxic communication pulls in users who engage by replying, retaliating, defending, or spectating. Overall, toxicity likely increases overall attentional on feed elements, including the ad.

These benefits of toxicity likely come to a halt at high toxicity. Sustained exposure to severely toxic content like

hate speech, harassment, dehumanizing language, produces aversion, lowered engagement, and withdrawal. Users report that they avoid participation for fear of harassment or shut down their accounts entirely; self-censorship and silencing are among the observed reactions (Lenhart et al. 2016). For instance, misogynistic hate speech reduces women's participation in political discourse (Carlson 2018). In gaming, toxicity decreases player engagement (Liu et al. 2025). Put differently, studies on content moderation find that reducing toxicity can increase participation (Matias 2019; Wise et al. 2006). Furthermore, the operationalization of toxicity that became the industry standard (Aridor et al. 2024) directly includes a heightened propensity to leave a discussion as a signal for its presence (Lees et al. 2022).

The traditional cue-utilization hypothesis is informative as well as it predicts a reversal in processing depth at high levels of arousal. Elevated arousal from negative affect narrows the range of environmental cues that receive processing resources (Easterbrook 1959). Under moderate toxicity, the attentional window still encompasses both toxic content and adjacent stimuli. Under high toxicity and thus arousal, attention concentrates on threat-relevant cues at the expense of peripheral stimuli (Martha and Sutherland, 2014; van Steenbergen et al. 2011). Users in a state of heightened threat vigilance prioritize monitoring toxic content and disengage from elements perceived as irrelevant to the arousal source. The ad — a commercially oriented, non-threatening element — thus loses attentional resources.

Ad effectiveness therefore depends on the overall levels of toxicity in the feed. At low-to-moderate doses, attentional toxicty likely exceeds aversion. At high doses, the relationship reverses. We therefore expect an inverted U: engagement increases with toxicity dose up to a threshold and declines beyond it.

> **H1:** Feed-level toxicity and ad engagement (ad dwell time, ad clickthrough rate, ad recall) follow an inverted-U relationship: ad engagement increases with the proportion of toxic content up to a moderate threshold and decreases at higher proportions.

Brand Safety

Spillover contamination from adjacent content is a central mechanism in brand safety research: consumers transfer negative associations from proximate content to a co-located brand, even absent any causal or semantic link between them (Grewal, Vana, and Stephen 2025). That is, when a brand is presented alongside unsafe content online, there is a spillover of the adjacent negativity that harms said brand. This spillover of contextual factors on co-located brands presented in the same context has been demonstrated in other settings, such as print and television (Bushman 2005; de Pelsmacker et al. 2002; Nian et al. 2021). Because toxicity is defined by its harm-oriented function, it constitutes unsafe content by definition, and spillover theory therefore predicts that toxicity in the content surrounding an ad harms brand attitudes.

The attention mechanism underlying H1, however, introduces level-dependence that prior spillover theory does not anticipate. Contamination requires encoding of the brand-context association, and encoding depth is a function of attentional resources allocated to the ad. At moderate toxicity (where H1 predicts the highest attention allocation) deeper processing of the ad occurs within a hostile environment, which should strengthen the brand-context association and amplify contamination. At high levels toxicity (where H1 predicts the withdrawal attentional resources toward threat-relevant cues) the ad receives fewer processing resources. This would weaken the association through which contamination operates even as the environment becomes more hostile. The net effect on brand attitudes is thus expected to resemble the attentional pattern in H1: contamination should be strongest at moderate toxicity and attenuate at high toxicity.

> **H2:** Toxicity in the content surrounding an ad harms brand attitudes, with the effect being strongest at moderate toxicity levels and attenuating at high toxicity.

Whether H2 holds in a naturalistic feed context is an open empirical question. Prior evidence for spillover contamination predominantly stems from studies that isolate brand-content pairings. For example, Lee et al. (2021, e.g., Study 1) presented participants with a single offensive social media post on animal abuse with a pet shop brand ad presented juxtaposed to the post. While such designs are certainly powerful to manipulate and capture psychological processes of brand safety concerns, they tend to neglect the feed structure, that is, the actual browsing in which consumers encounter brand safety incidents. This may affect the predicted effects in two ways. First, these studies concentrate processing resources on the brand-content juxtaposition and are likely to induce an overly analytical mindset [Pham (2013); Sawyer 1975]. This suggests that prior estimates may overstate what occurs under feed conditions. Second, prior designs operationalize local adjacency rather than feed-level toxicity. What typically varies is the content directly neighboring the ad, not the toxicity of the entire feed. In contrast, the present study varies feed-level toxicity independently of adjacent-post toxicity. Doing so, we intend to estimate a quantity that prior research has not directly addressed. How strongly spillover contamination operates through feed-level toxicity, therefore, remains an open question.

Motivating Evidence

Research on social media platforms shows that toxic content drives *engagement*: it attracts more reactions, shares, and time spent than civil alternatives (Beknazar-Yuzbashev et al. 2022; Aridor et al. 2024; Beknazar-Yuzbashev, Jiménez-Durán, and Stalinski 2024). We provide evidence that toxicity also drives *attention* specifically, using 12,473 A/B tests from the *Upworthy Research Archive* (Matias et al. 2021) that cover 53,755 news headlines that were displayed 205.8M times.

In this setting, each A/B test randomly assigned website visitors to one of several headlines referring to the same article and recorded clicks on each headline. Unlike likes or comments, these 2.8M clicks were not visible to others. The clicks in this dataset therefore reflect attentional responses rather than social behavior, which is precisely the channel our theoretical argument concerns.

The recorded click-through rates (CTRs) range from 0.000% to 14.891%. The average CTR across all headlines is 1.394% and the median click rate is 1.067%. As the distribution among CTRs is right-skewed, only a small proportion of news stories are associated with a high CTR.

Toxic news headlines, defined as those exceeding a Perspective API score of 0.6, achieve higher click-through rates than civil alternatives ($Median_{CTR}^{Civil} = 1.061\%$; $Median_{CTR}^{Toxic} = 1.401\%$). Figure 1 provides model-free support for this claim: the distribution of click-through rates is shifted rightward for toxic headlines across the full range of observed values. A fixed-effects regression that compares headlines which refer to the same article quantifies the difference: the estimated uplift corresponds to 6.408% of the median click-through rate, and it holds after controlling for headline length, readability, and negative emotional valence.

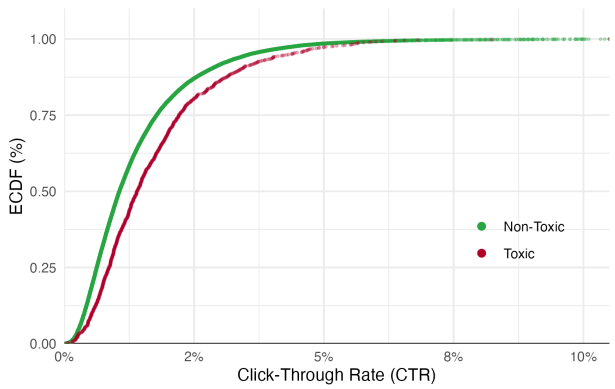

Figure 1. Empirical cumulative distribution functions of toxic vs. non-toxic headlines

These results suggest that toxicity drives attention in a setting where article content is held constant and social motives are absent. Importantly, the Upworthy data represent an unusually clean source of field evidence for this purpose as the A/B tests were designed by the platform itself: visitors were randomly assigned to headline variants, one experiment ran at a time, and commercial incentives ensured careful implementation. Such data are rare. Even so, three limitations prevail. First, the data contain no information about the feed structure in which each headline was embedded, which is why we cannot derive a dose-response relationship between toxicity and attention. Furthermore, this study operationalizes clicks on the stimulus itself. It does therefore not offer evidence of excitation transfer from toxic content to adjacent, non-toxic content. Finally, it does not provide evidence on downstream consequences such as brand attitudes.

Addressing these limitations with additional field data that is typically available to marketing scholars is difficult, if not impossible. This is because online platform studies and observational studies both suffer from algorithmic interference: platform algorithms determine which users see which content in ways that are unobservable to researchers, making it difficult to isolate the effect of feed composition on any outcome (Xu, Zhang, and Zhou 2020; Johnson 2023; Braun and Schwartz 2025). Furthermore, neither paradigm allows researchers to manipulate or even observe the context surrounding a focal ad, which is required to study brand safety (Roggenkamp, Boegershausen, and Hildebrand 2025). In what follows, we therefore turn our attention to an experimental design.

Experimental Evidence

To address the three limitations of the news headline study, we designed a Digital In-Context Experiment to embed a sponsored post within a feed whose toxicity dose is experimentally varied. This enables a clean operationalization as well as more ecologically valid stimuli and measures compared to typical vignette experiments (Roggenkamp, Boegershausen, and Hildebrand 2025).

Procedure

Participants were recruited via Prolific and directed to an oTree-based web application. After providing informed consent, participants read a brief scenario framing the task: they had recently come across the latest *Avatar: Fire and Ash* film and were opening a microblogging feed to see what others were saying about it. They were instructed to browse at whatever pace and depth felt natural and to click a "Done browsing" button when they felt they had seen enough.

Participants then viewed a simulated microblogging feed consisting of one sponsored and 25 organic posts. Each feed was composed by first classifying all posts in the stimulus corpus as toxic (Perspective API score ≥ 0.6) or non-toxic (score < 0.6), yielding two sampling pools. Each participant was assigned a toxicity dose $d \in [0, 1]$, representing the target proportion of toxic organic posts in their feed. For a participant assigned dose $d = \frac{3}{25}$, for

instance, the feed was constructed by sampling three organic posts without replacement from the toxic pool and the 22 organic remaining posts from the non-toxic pool, then randomly shuffling the resulting sequence. Because each participant draws a fresh independent sample, participants assigned the same dose see different posts and sequences to ensure that idiosyncratic post-level features are uncorrelated with the dose across participants.

One post in each feed was a sponsored post (the focal advertisement) which was placed at a fixed (i.e., the 14th) scroll position. The sponsored post was identical across all participants and all dose conditions. Because post order within the feed was randomized, the posts appearing immediately above and below the sponsored post varied randomly across participants. The local toxicity of the ad's immediate context therefore co-varies with the feed-level dose in expectation but is not deterministically controlled, which we address in the analysis by including the toxicity of adjacent posts as a covariate..?

Participant behavior during feed exposure was recorded unobtrusively. A viewport tracker logged the duration each post was visible in the browser window (dwell time) to provide a proxy for in(attention). Participants could also like posts and compose reply text, which were recorded as indicators of active engagement. Total time from loading the feed in the browser to proceeding to the next page was recorded as overall time on feed. We also recorded paradata, such as the device type and screen resolution to allow for covariate adjustment for differences in feed rendering across devices. After browsing, participants were automatically redirected to a Qualtrics survey to elicit ad recall and brand attitudes.

Pilot Data

Lorem ipsum

Sampling Plan

Lorem ipsum

Analysis Plan

H1: Feed toxicity and ad engagement

H1 predicts that feed-level toxicity and ad engagement follow an inverted-U relationship. Outcomes are ordered by their theoretical proximity to the attention mechanism and their statistical sensitivity, establishing a pre-registered primary/secondary hierarchy to control the family-wise Type I error rate.

Ad dwell time (log-transformed) is the primary outcome. It is the most direct unobtrusive measure of attention allocated to the sponsored post, recorded continuously by the viewport tracker without requiring any active response from participants. The sample size is powered for this outcome (see **?@sec-samplesize**). H1 is confirmed or disconfirmed on the basis of this outcome alone.

Unaided ad recall is the first secondary outcome. It captures the downstream consequence of sustained attention — whether exposure was sufficient to encode the brand into memory — and is stricter and less susceptible to guessing than aided recall.

Aided ad recall is the second secondary outcome, reported as a robustness check when unaided recall is sparse.

Lottery signup (binary) is the third secondary outcome. Although behavioral, it is influenced by factors beyond attention (e.g., individual propensity to enter lotteries) and is relatively sparse, limiting power. It is reported for completeness but not treated as confirmatory.

Time on feed (log-transformed) is reported as a feed-level mechanism check rather than a test of H1. It addresses whether feed-level toxicity affects overall browsing duration — a precondition for the attention mechanism — but is not an ad effectiveness measure.

All secondary outcomes are tested at $\alpha = .05$ using the same two-lines procedure as the primary outcome but are not the basis for confirming or rejecting H1.

We follow the two-lines procedure proposed by Simonsohn (2018). For each outcome, we 1. fit a cubic spline to estimate the location of the peak, 2. apply the Robin Hood algorithm to determine the break point that maximizes power by allocating observations toward the weaker of the two slopes, and 3. estimate an interrupted regression with one slope below and one slope above the break point. H1 is fully supported if both slopes (denoted b_1 and b_2 in Table 2) pass one-sided tests ($\alpha = .05$) in the predicted directions: $b_1 > 0$ below the break point and $b_2 < 0$ above it. Because the two-lines test requires a true sign reversal, a positive-then-flat trajectory — in which engagement increases and then saturates without declining — would not satisfy the strict criterion but is still broadly consistent with the arousal mechanism and is treated as partial support.

To capture this possibility, we additionally estimate a quadratic regression of the outcome on `feed_toxicity` and `feed_toxicity`^{2}. A significant negative quadratic coefficient alongside the plateau pattern in the two-lines test is interpreted as partial support for H1: the data show the predicted initial increase and a curvilinear compression, even if the second slope does not turn significantly negative. This secondary test is reported for completeness but, following Simonsohn (2018), is not treated as primary evidence given the known tendency of quadratic regression to produce false positives. The full decision rules are in Table 2.

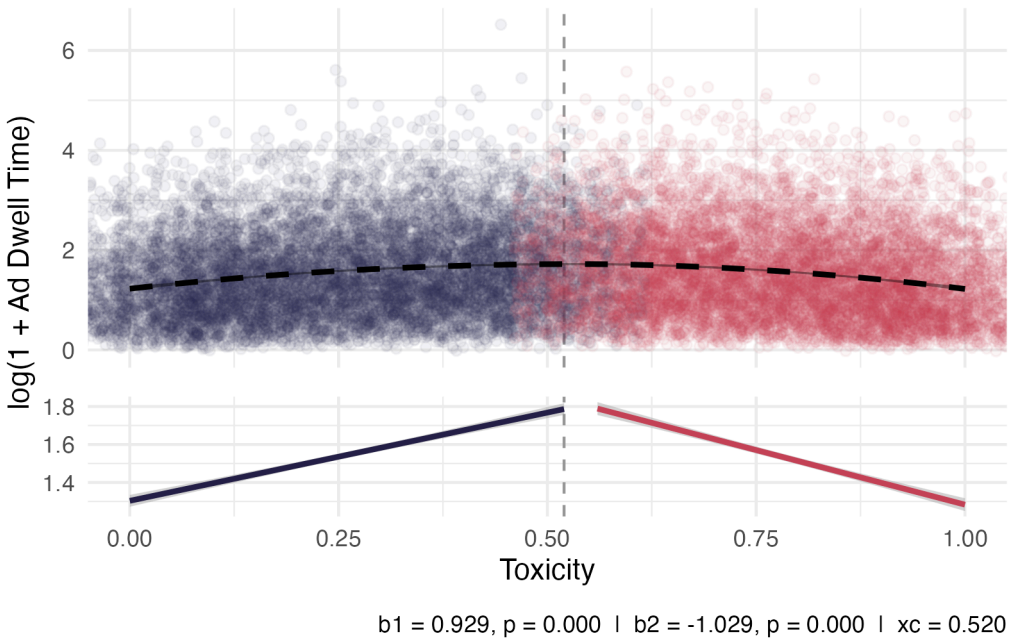

Figure 2. Two-lines test: feed toxicity and ad dwell time. Upper panel shows raw data with LOESS smoother; lower panel shows the two fitted regression lines. Break point determined by the Robin Hood procedure (Simonsohn 2018).

Table 2. Pre-registered decision rules for H1. Quadratic term column is only evaluated when the two-lines test does not yield opposite significant slopes.

Two-lines result	Quadratic term	Interpretation
$b_1 > 0^*$, $b_2 < 0^*$	–	**Support H1** (inverted-U confirmed)
$b_1 > 0^*$, b_2 n.s.	$< 0^*$	**Partial support H1** (plateau: engagement increases and saturates but does not invert)
$b_1 > 0^*$, b_2 n.s.	n.s.	Reject H1 (monotonic increase)
$b_1 > 0^*$, $b_2 > 0^*$	–	Reject H1 (monotonic increase)
b_1 n.s. or b_2 n.s.	–	Reject H1 (no detectable relationship)

H2: Feed toxicity and brand attitude

H2 predicts an inverted-U relationship between feed-level toxicity and brand attitude, with contamination strongest at moderate toxicity and attenuating at high toxicity. We apply the same two-lines procedure and quadratic fallback as for H1 (see Table 2). The primary test is whether both slopes are significant with opposite signs; partial support is declared if the first slope is positive and significant and the quadratic term is significantly negative.

We compute a composite brand attitude score as the mean of the six focal brand items (`gfb_1` to `gfb_6`). If neither the two-lines test nor the quadratic term is significant, we conclude that feed-level toxicity does not affect brand attitude in a detectable way, consistent with the attenuation argument in the brand safety section above.

Sample Size Justification

The two-lines test has no closed-form power formula, so we estimate power by simulation following Lakens (2022). For each candidate sample size N, we

1. generate a dataset under one of two assumed inverted-U data-generating processes,
2. apply the two-lines test,
3. flag a success when both slopes pass a one-sided test ($\alpha = .05$; $b_1 > 0$ and $b_2 < 0$), and
4. repeat the procedure 1,000 times.

Power is the proportion of successful replications. A key concept throughout is the **inverted-U peak**: under the assumed model $\mathbb{E}[Y \mid x] = \beta_0 + \beta_1 \cdot x(1-x)$, the function $x(1-x)$ reaches its maximum of 0.25 at $x = 0.5$ (a feed that is exactly half toxic). The peak outcome is

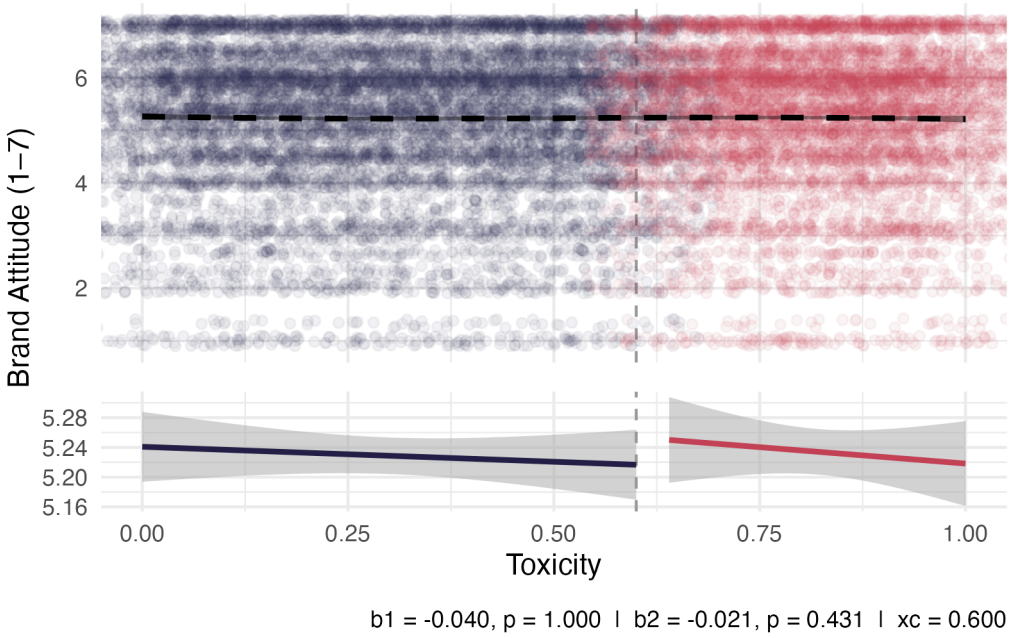

Figure 3. Two-lines test: feed toxicity and brand attitude.

therefore $\beta_0 + 0.25\,\beta_1$, and $0.25\,\beta_1$ is the peak uplift — how much higher the expected outcome is at 50% toxicity compared to a fully clean or fully toxic feed. Effect sizes are expressed as Cohen's d at this peak: $d = 0.25\,\beta_1/\sigma$, or equivalently $\beta_1 = 4\,d\,\sigma$.

We report power separately for two outcomes — ad dwell time and brand attitude — using this common d scale. The power curves (Figure 4) indicate that detecting a small effect ($d = 0.2$) with 80% power requiresapproximately $N = 4,000$ participants; we adopt this as our recruitment target. The two-lines test is more conservative than a simple quadratic regression, so this sample size provides at minimum adequate power for the preregistered primary test.

Ad dwell time. Because ad dwell time is heavily right-skewed, we analyze it on the $\log(1 + \text{seconds})$ scale, with σ estimated from the pilot. The pilot baseline is 2.2 s (median dwell time at the tails, i.e., in a fully clean or fully toxic feed). The peak — a 50% toxic feed — raises median dwell time to 3.6 s under a medium effect ($d = 0.5$), an absolute increase of 1.4 s. Under a small effect ($d = 0.2$), the peak median is 2.7 s.

Brand attitude. We apply the same d = 0.5 / d = 0.2 convention. The pilot mean is 5.24 on a 1–7 scale (SD: 1.38). A medium effect raises peak-condition attitude by 0.69 points; a small effect by 0.28 points.

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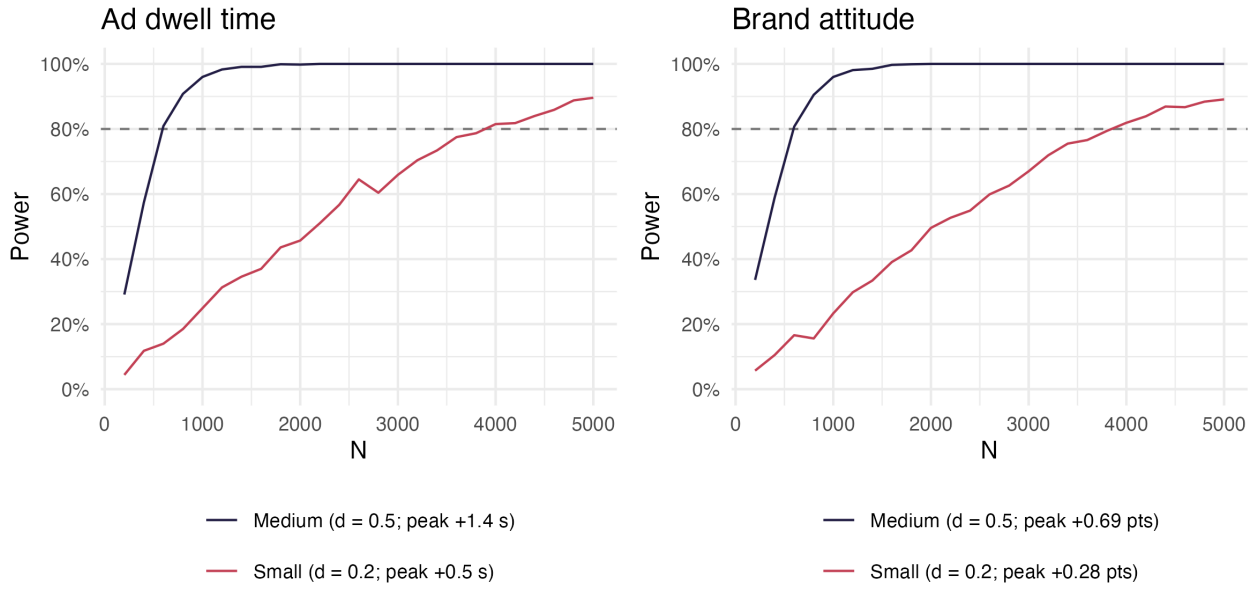

Figure 4. Power curves for the two-lines test on ad dwell time (left) and brand attitude (right). Effect sizes are expressed as Cohen's d at the inverted-U peak (d = peak uplift / pilot SD). Dashed line marks 80% power.

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